

0.1 Introduction

I have always been a little fascinated by flight crews: the idea of being paid to fly somewhere and then getting to travel once you land (at least that is what I imagined the job to be). That curiosity is what drew me to this report, which aims to reproduce and potentially improve upon the results of the paper “Individual scheduling approach for multi-class airline cabin crew with manpower requirement heterogeneity”¹ [1]. This report focuses on the integer programming aspects of the paper.

0.1.1 Background Problem

The paper tackles the cabin crew pairing problem (CPP): building legal sequences of flights (pairings) that start and end at a home base and cover every flight, while minimising cost. Crew cost is an airline’s second-largest expense after fuel, so small scheduling gains matter. Cockpit crews are qualified for one aircraft type with fixed requirements and are scheduled as teams. Cabin crews are different: they are split into multiple classes, are cross-qualified across aircraft types, and the number of each class required varies by aircraft, cabin layout, and flight. This heterogeneity is the core difficulty.

Modelling cabin crews as teams is wasteful here. Since a team flies together, its size must meet the maximum requirement across all flights on the pairing, leaving surplus crew idle on lighter flights (manpower waste). Scheduling crew individually avoids this and raises utilisation. The paper also formalises Controlled Crew Substitution (CCS): when flight fluctuation leaves a class short, a crew member of another class temporarily covers the duty, so a flight is fine as long as total crew meets total demand. Combining these ideas gives the proposed MICCPP-ACCS model, which schedules each crew individually, adds per-class availability limits, and embeds CCS.

0.1.2 Original Paper’s methodology

The paper’s contribution is as much conceptual as computational: it argues that cabin crews should be scheduled as individuals rather than as fixed teams, and backs this with three integer programming models. The traditional model (TCCPP) is the team-based baseline the literature has long used. The proposed model (MICCPP-ACCS) schedules each crew member of each class individually, caps how many crew each class has available, and embeds Controlled Crew Substitution so a surplus class can temporarily cover a short one. A stripped-down variant (MICCPP-A) drops substitution and is used mainly to derive benchmarks. In every case the objective is driven by time-away-from-base, with substitution and extra hiring penalised heavily enough that the model only reaches for them when genuinely short of crew.

From these models the paper derives manpower-requirement benchmarks for a schedule and uses them to sort any availability situation into a handful of scenarios, showing when substitution alone suffices and when extra crew must be hired. The models are solved with a column-generation heuristic: pairings are generated on demand by solving shortest-path subproblems over a duty network, and integer solutions are recovered by running a MIP over only the generated columns, with a genetic algorithm standing in for the largest instances. Experiments on a Hong Kong–Singapore route and on large synthetic instances report that the individual approach removes idle-crew waste and lowers cost relative to the team baseline, with substitution further cutting the need for extra hires.

0.1.3 Limitations of the original models

Three modelling choices stand out as limitations. First, the formulation plans around a single home base (Hong Kong): availability is just a cap on pairings per class, and any shortage is filled by “extra” crew that the model can effectively import from anywhere at a fixed penalty. This sidesteps the real geography of crew bases and positioning, so the availability constraint is a rough proxy rather than a true resourcing model. Second, the model lets any class substitute any other class in either direction. If substitution is unrestricted, it is unclear what the class distinction is buying; in practice higher classes cover lower ones, not the reverse, and free two-way substitution risks overstating the flexibility CCS delivers. Third, cost is approximated purely by time-away-from-base, treating every minute of a pairing as equal. In reality the segments of a duty cost differently — active flight time, ground waiting

¹X. Wen, S.-H. Chung, P. Ji, and J.-B. Sheu, *Transportation Research Part E*, vol. 163, 102763, 2022.

(sit) time, and deadhead repositioning are not interchangeable — so a single TAFB figure can misrank pairings an airline would price very differently.

0.2 Problem Setups

The instances are rebuilt from public data so the study is reproducible without the paper’s confidential figures.

0.2.1 Flight Data Source

Flights come from the BTS On-Time Performance dataset (tail number, origin / destination, scheduled times, delay, cancellation). Aircraft attributes are taken from the FAA registry and attached by a left join on the tail number, giving each flight f a seat count s_f . The seat count resolves from the per-tail registry value, falling back to a model→seats table when missing.

The minimum cabin crew m_f is the regulatory floor (14 CFR 121.391):

$$m_f = \begin{cases} 1 & \text{if } s_f \leq 50 \\ 2 & \text{if } 50 < s_f \leq 100 \\ 2 + \lceil (s_f - 100)/50 \rceil & \text{if } s_f > 100 \end{cases}$$

which yields a heterogeneous per-flight requirement, as in the paper, but from open data. (See the appendix for one enriched record.)

0.2.2 Crew Base Allocation

The paper’s single base is replaced by one base per airport the airline both departs from and arrives at. With $\mathcal{O}$ the set of origin airports and $\mathcal{D}$ the set of destinations,

$$B = \mathcal{O} \cap \mathcal{D}.$$

Each base $p \in B$ is sized from two signals. Let d_f be flight duration and $u \in (0, 1]$ a utilisation discount over an H -day horizon; the per-crew duty capacity and the duration-weighted demand (with a slack factor 1.8) are

$$\tau_{\text{duty}} = 8 \cdot 60 \cdot H \cdot u, \quad n_p^{\text{dem}} = \left\lceil \frac{1.8}{\tau_{\text{duty}}} \sum_{f: \text{orig}(f)=p} m_f d_f \right\rceil.$$

Writing $L_p(t)$ for the simultaneous crew load at p (a sweep adding $+m_f$ at each departure and $-m_f$ at each arrival), the peak signal and final count are

$$n_p^{\text{peak}} = \left\lceil 1.8 \cdot \max_t L_p(t) \right\rceil, \quad n_p = \text{clip}\left(\text{round}\left(\mathcal{N}\left(\mu_p, (0.1\mu_p)^2\right)\right), n_{\min}, n_{\max}\right),$$

where $\mu_p = \max(n_p^{\text{dem}}, n_p^{\text{peak}}, n_{\min})$ and the Gaussian draw mimics real over-provisioning. Availability is thus demand-proportional and location-aware, rather than an unlimited pool imported from anywhere.

0.2.3 Assumptions

The upstream stages (scheduling, fleet assignment, aircraft routing) are fixed inputs; the planning horizon is $T = 1$ week; every pairing starts and ends at its crew member’s base; and crew are interchangeable within a class, subject to the substitution rules. Disruption response and the early alternative solvers are out of scope here and discussed later.

0.2.4 Rolling Horizon

The dataset spans more than one horizon, so planning advances in successive windows of length $L = T$ with step Δ :

$$W_k = [k\Delta, k\Delta + L), \quad k = 0, 1, 2, \dots$$

Each W_k is solved as a static instance and its assignments committed before advancing to W_{k+1} , keeping every solve tractable and leaving a seam where later disruption handling can attach.

0.3 Appendix: Enriched Record Format

One flight after the BTS→FAA join and crew derivation. Times are minutes since local midnight; **duration** is scheduled elapsed time.

```
{
  "flight_id":      "AA0001_2025-01-01",
  "carrier":        "AA",
  "flight_number":  1,
  "date":           "2025-01-01",
  "origin":          "JFK",
  "dest":            "LAX",
  "dep_min":         419,
  "arr_min":         620,
  "duration":        381,
  "tail_num":        "N104NN",
  "aircraft": {
    "manufacturer": "Airbus",
    "model":         "A321-231",
    "seats":          185,
    "seat_source":    "faa_registry"
  },
  "min_crew":        4,
  "cancelled":        false
}
```

A crew member carries only an id and its home base; its class is assigned by the model.

```
{ "id": 0, "base": "JFK", "class": "general" }
```